MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code: BBA(HM) 202 | Medical Record Science I | UPID: 2000082

Time: 3 Hours | Full Marks: 70 | Sem 2 — 2024-2025

Complete Answer Key

GROUP-A | Very Short Answer Type | 1 × 10 = 10 Marks

Answer any **ten** of the following in 1–2 lines each.

Question (i)

What is the full form of MR in healthcare?

MR = Medical Record

A Medical Record is the complete written account of a patient's health — their history, diagnosis, treatment, and outcome — kept safely in the hospital.

Question (ii)

Name any two types of medical records.

Type	Meaning
Inpatient Record	Record of a patient who is admitted and stays in the hospital (e.g., surgery case sheets).
Outpatient Record	Record of a patient who visits the hospital for treatment but goes home the same day (e.g., OPD card).

Question (iii)

Name one method used for preserving medical records.

Microfilming
In this method, paper records are photographed and stored on a small film roll. This saves a lot of storage space and keeps records safe for 100+ years without damage.

Question (iv)

Mention one role of an MR technician.

Medical Coding (ICD Coding)
An MR technician converts the doctor's diagnosis and procedures into standard codes using the ICD (International Classification of Diseases) system. These codes are used for billing, insurance, and hospital statistics.

Question (v)

What is meant by preservation of medical records?

Definition: Preservation of medical records means protecting and maintaining patient records from damage, decay, theft, or loss — so they remain available and readable for as long as legally required.

Methods include: microfilming, digitization, fire-proof storage, and climate-controlled rooms.

Question (vi)

Who is responsible for managing the Medical Records Department?

The **Medical Record Officer (MRO)** — also called the **Health Information Manager (HIM)** — is the head of the MRD. They are responsible for managing all staff, systems, policies, and legal compliance in the department.

Question (vii)

Define medical audit.

Medical Audit is a systematic review of patient care by checking medical records against set standards, to find out if the treatment given was correct and to improve quality of care.

💡 Simple way to remember: "Check the care → Compare to standards → Improve where needed."

Question (viii)

Name any one legal document maintained in medical records.

Informed Consent Form

This is a document signed by the patient (or guardian) before any procedure or surgery. It shows the patient understood the risks and agreed to the treatment. It protects the hospital legally.

Question (ix)

What is one characteristic of a good medical record?

Accuracy

A good medical record must contain correct and truthful information — the right patient name, correct diagnosis, right dates, and accurate treatment details — with no errors or guesses.

Question (x)

Define retention of medical records.

Retention means keeping medical records stored safely for a fixed period of time as required by law or hospital policy — before they can be destroyed or archived.

Example: Adult records — 5 to 10 years; Minors — till they turn 18 + extra years; MLC records — 10+ years.

Question (xi)

What is a medico-legal case?

Medico-Legal Case (MLC) is any case where a patient's injury or illness has legal importance — meaning the police, court, or law is involved.

Examples: Road accidents, physical assault, poisoning, rape, suspicious death, burns.

Question (xii)

Name one type of record involved in medico-legal cases.

Post-Mortem Report (Autopsy Report)
This report is prepared by a forensic doctor after examining a dead body. It states the exact cause and manner of death and is used as important evidence in court.

GROUP-B | Short Answer Type | 5 × 3 = 15 Marks | Answer any THREE

Question 2

Describe the different types of medical records with examples. 5 Marks

Medical records are documents that contain all information about a patient's health. They are of different types based on the patient's situation:

Type	Who it is for	What it contains	Example
Inpatient Record	Admitted patients	Admission notes, progress notes, operation notes, discharge summary	Surgery patient's case sheet
Outpatient Record	OPD (visiting) patients	Doctor's consultation notes, prescriptions, test reports	OPD card / folder
Emergency Record	Casualty/Emergency patients	Triage notes, emergency treatment given, MLC details	Accident & Emergency register
Diagnostic Record	All patients needing tests	Lab reports, X-rays, ECG, MRI, scan reports	Blood test report

Medico-Legal Record	MLC patients	Wound certificate, post-mortem report, MLC register entry	MLC register
Administrative Record	All patients	Birth/death certificates, transfer letters, consent forms	Death certificate

💡 **Tip to remember:** In-Out-Emergency-Diagnostic-Medico-Admin → **I O E D M A**

Question 3

Explain the importance of medical records in legal cases. **5 Marks**

Medical records play a very important role in legal situations. Here is how:

#	Legal Importance	Explanation
1	**Evidence in Court**	Records show exactly what treatment was given. If a doctor is accused of negligence, the record is proof of correct care.
2	**Medico-Legal Cases**	In accidents, assaults, or poisoning cases, the MLC record becomes the official document used by police and court.
3	**Insurance Claims**	Insurance companies use medical records to verify the illness or injury and process claims correctly.
4	**Consent Documentation**	Signed consent forms protect the hospital and doctor from legal action if the patient later complains about a procedure.
5	**Death Investigation**	In suspicious deaths, hospital records and post-mortem reports help determine if it was natural or unnatural death.
6	**Patient's Right to Access**	By law, patients have the right to get copies of their own records for any legal or personal purpose.
7	**Doctor's Accountability**	Records prove what decision was taken, when, and why — making doctors accountable for their actions.

Key Statement: "A medical record is the doctor's best friend in court and the hospital's protection against false claims."

Question 4

Describe how MRD handles medico-legal documents. 5 Marks

When a Medico-Legal Case (MLC) comes to the hospital, the MRD follows a special step-by-step process:

① Identification	Patient is identified as MLC at Emergency/OPD entry. A special MLC tag is attached to the case sheet.
↓	
② Separate Registration	MLC is entered in a dedicated MLC Register with a unique MLC number, date, time, and nature of injury.
↓	
③ Police Intimation	Police are immediately informed (by phone and in writing) about the MLC as required by law.
↓	
④ Secure Storage	MLC files are kept in locked, restricted-access areas. Only authorized persons can access them.
↓	
⑤ Chain of Custody	Any document given to police or court is recorded — who received it, when, and for what purpose.
↓	
⑥ Long Retention	MLC records are kept for 10+ years because legal cases may come up years later.

💡 MLC records are the most sensitive records in any hospital — accuracy and confidentiality are extremely important.

Question 5

Write a short note on assembling and deficiency checking of medical records. 5 Marks

📋 ASSEMBLING

What: After a patient is discharged, all loose papers are collected and arranged in a fixed order.

Standard Order:

- Face Sheet (patient details)
- History & Physical Examination
- Doctor's Orders

🔍 DEFICIENCY CHECKING

What: Each assembled record is checked for missing or incomplete parts.

Common Deficiencies Found:

- Unsigned discharge summary
- Missing diagnosis / ICD code
- Incomplete history & examination

- Progress Notes
- Lab / Radiology Reports
- Consent Forms
- Discharge Summary

- Missing consent form
- Illegible (unreadable) entries

Patient Discharged	All loose papers collected from the ward
↓ **Assembling**	Papers arranged in standard hospital order
↓ **Deficiency Check**	Record checked for missing signatures, reports, and entries
↓ **Deficiency Slip**	Slip sent to concerned doctor to complete missing entries
↓ **Filing**	Only complete records are filed permanently in MRD

Question 6

Briefly describe the organizational structure of a Medical Records Department. **5 Marks**

The MRD has a clear hierarchy with different staff handling different responsibilities:

Hospital Administrator

Medical Record Officer (MRO) / HIM Manager
Head of MRD — Overall management, policy, legal compliance

MR Technicians
ICD Coding, Indexing, Retrieval

MR Assistants
Filing, Assembling, Deficiency Check

Statistics Section
Hospital statistics & reports

Birth & Death Registration | Medico-Legal Section (MLC) | Coding Section (ICD) | Record Storage & Retrieval

Remember: The MRD works alongside ALL departments in the hospital — it is the backbone of hospital information management.

GROUP-C | Long Answer Type | 15 × 3 = 45 Marks | Answer any THREE

Question 7

Explain various methods used in the preservation of medical records, both manual and electronic.

15 Marks

Preservation of Medical Records means protecting patient records from damage, decay, loss, or unauthorized access so they remain available for as long as needed — for patient care, legal purposes, and research.

📁 MANUAL METHODS

- **Microfilming:** Records are photographed onto a small film roll. Very space-saving and can last 100+ years. Used in large hospitals.
- **Microfiching:** Similar to microfilming but stored on flat plastic cards (microfiche). Easy to view on a reader machine.
- **Lamination:** Important single papers are covered in plastic film to protect from moisture, tearing, and handling damage.
- **Acid-free folders:** Special folders that do not release chemicals that damage paper over time. Used for long-term storage.
- **Fire-proof cabinets:** Metal cabinets that protect records from fire accidents.
- **Climate-controlled rooms:** Records stored at controlled temperature and humidity to prevent yellowing, fungus, and decay.
- **Pest control:** Regular spraying to prevent insects (white ants, cockroaches) from destroying paper records.

💻 ELECTRONIC METHODS

- **EHR (Electronic Health Records):** All patient information is entered directly into a computer system — no paper needed. Most modern and secure method.
- **Scanning & Digitization:** Old paper records are scanned and saved as PDF/image files on computer servers.
- **Cloud Storage:** Digital records stored on secure internet servers. Can be accessed from anywhere anytime.
- **RAID Systems:** (Redundant Array of Independent Disks) — Multiple hard drives store the same data so if one fails, data is not lost.
- **Optical Discs (CD/DVD/Blu-ray):** Records saved on discs and stored in a library. Good for long-term archiving.
- **Encrypted Backup:** Automatic backup of data to a secondary location with encryption (password protection) to prevent hacking.
- **Disaster Recovery Systems:** System that restores all data automatically if the main server crashes or is destroyed.

Comparison Table:

Feature	Manual Methods	Electronic Methods
Storage space	Needs large physical space	Very compact (server/cloud)
Retrieval speed	Slow (manual search)	Instant (search by keyword)
Security	Lock & key, fire-proof cabinets	Password, encryption, firewall
Cost	Low initial cost	Higher setup cost, lower running cost
Risk of damage	Fire, flood, pests, decay	Hacking, power failure (mitigated by backup)
Best use	Rural/older hospitals	Modern, large hospitals

Conclusion: The best approach is a combination of both — keeping electronic copies of all records while maintaining secure physical copies of the most critical documents.

Question 8(a)

Elaborate on the types, process, and outcomes of medical audits. Why are they essential in hospitals?

10 Marks

Medical Audit is a quality improvement process where the care given to patients is checked against set standards — to find problems and fix them to improve patient safety and treatment quality.

Types of Medical Audit:

Type	When done	Purpose
Prospective Audit	Before treatment starts	Plan the right care in advance
Concurrent Audit	While patient is being treated	Correct problems immediately
Retrospective Audit	After patient is discharged	Review patterns, most common type
Internal Audit	Anytime	Done by hospital's own team
External Audit	Scheduled by accreditation body	Done by NABH/JCI or outside experts

The Audit Cycle (Process) — 7 Steps:

Choose Topic → Step 2 Set Standards → Step 3 Collect Data → Step 4 Compare Results

Step 5 Find Gaps → Step 6 Make Changes → Re-Audit

💡 This is called the **Audit Cycle** — after Step 7, you go back to Step 1 again for continuous improvement.

Outcomes of Medical Audit:

Outcome	Benefit
Improved quality of care	Patients receive better, standardized treatment
Fewer medical errors	Wrong treatments or missed diagnoses are identified and fixed
Better documentation	Doctors become more careful and complete in their record writing
Accountability	Clinicians are responsible for their decisions
Hospital accreditation	Meets NABH/JCI standards for quality certification

Why Essential? Medical audits make hospitals safer places for patients. They find mistakes before they become big problems, encourage doctors to follow best practices, and help hospitals earn trust and accreditation.

Question 8(b)

What are the guidelines for determining staffing levels and requirements in the MRD? 5 Marks

The number of staff needed in an MRD depends on several factors:

#	Factor	How it affects staffing
1	**Number of Beds**	More beds = more patients = more records = more staff needed. General ratio: **1 MR technician per 50–100 beds**
2	**Daily Patient Load**	High OPD and IPD volume means more record creation, assembly, and filing work.
3	**Type of Hospital**	Teaching hospitals need more staff for research, coding, and training activities.

4	**Services Available**	Hospitals with surgery, ICU, oncology generate more complex records needing expert coders.
5	**Turnaround Time**	If records must be ready in 24 hours, more staff is needed to meet the deadline.
6	**Level of Computerization**	Fully digital systems reduce manual work, so fewer staff are needed for routine tasks.

💡 **Standard Ratio:** 1 MR Technician per 50–100 beds | 1 MRO per department | Specialized coders for ICU, surgery, and MLC sections.

Question 9

How do MRD personnel ensure legal compliance and protect patient rights in the handling of records? **15 Marks**

MRD personnel have a dual responsibility: (1) following the **law** in how records are managed, and (2) protecting the **rights of every patient** whose information is stored.

⚖️ LEGAL COMPLIANCE

- **Retention Policy:** Records kept for legally required duration (adults: 5–10 yrs; minors: till 18 + extra yrs; MLC: 10+ yrs).
- **Authorized Access Only:** Records given only to treating doctors, patient, court order, or insurance with valid proof.
- **MLC Protocol:** All medico-legal cases are registered, police are informed, and strict documentation is maintained.
- **Accurate ICD Coding:** Correct codes ensure proper billing and avoid insurance fraud — a legal requirement.
- **Proper Consent Filing:** All consent forms filed correctly as they serve as legal protection for the hospital.
- **Birth & Death Registration:** Issuing official government documents as per law.

🛡️ PROTECTING PATIENT RIGHTS

- **Confidentiality:** Patient information is never shared without consent. Staff are bound by an oath of secrecy.
- **Right to Access:** Patients can legally ask for a copy of their own records anytime.
- **Right to Correction:** If there is a wrong entry, the patient can request it to be corrected.
- **Data Security:** Digital records are password-protected, encrypted, and access-controlled.
- **Audit Trails:** Every time a record is accessed digitally, a log is saved — who accessed it, when, and why.
- **Staff Training:** Regular training on privacy laws, ethics, and data protection policies.

Flow of Record Release (Legal Process):

Request Received	From patient, police, court, insurance, or doctor
↓	
Verify Authority	Check if the person requesting has valid authorization
↓	
MRO Approval	Medical Record Officer approves the release
↓	
Record Provided	Copy given — original stays in MRD always
↓	
Entry in Release Register	Who received it, date, purpose — all documented

Key Rule: The ORIGINAL record NEVER leaves the MRD. Only certified copies are given out.

Question 10

How can the Medical Records Department ensure the accuracy, confidentiality, and timely availability of patient records while adapting to digital transformation and increasing patient volumes?

15 Marks

As hospitals grow bigger and move to digital systems, the MRD must handle more records, faster, more securely — without any errors. This requires smart systems and trained staff.

Challenge	Strategy / Solution	Tool Used
Ensuring Accuracy	• EHR systems with auto-validation (no wrong formats accepted) • Regular deficiency checking after each discharge • Standardized ICD-10/11 coding • Periodic record audits • Staff training on data entry	EHR Software, ICD Coder, Audit Committee
Ensuring Confidentiality	• Role-based access (nurse sees only nursing notes, not billing)	Firewall, SSL encryption, Access Control Software

	• End-to-end data encryption • Multi-factor login authentication • Regular cybersecurity checks • Strict staff confidentiality policy	
Timely Availability	• Cloud systems allow instant access from anywhere • Barcode/RFID tagging for fast physical retrieval • 24/7 digital access for emergency teams • Disaster recovery ensures no downtime • Automated report generation	Cloud Server, Barcode System, Disaster Recovery
Managing More Patients	• Scalable cloud infrastructure (capacity grows automatically) • Automation of routine tasks (appointment records, discharge summaries) • Workflow management to divide workload among staff • Hiring and training more MRD staff as volume grows	HIS (Hospital Information System), Workflow Software

Digital Transformation Journey of MRD:

Paper Records (Manual) → **Microfilming** (Semi-manual) → **Scanned Digital** (Hybrid) → (Fully Digital) → **Cloud + AI** (Future)

Conclusion: The MRD of the future must be fast, secure, accurate, and paperless — using technology to handle growing patient volumes without compromising privacy or quality.

Question 11

Discuss the role of medical records in healthcare delivery. Include a flow chart of their function.

15 Marks

Medical records are the backbone of healthcare. Every decision a doctor makes, every treatment given, every medicine prescribed — all of it is recorded. Without medical records, healthcare delivery would be incomplete, unsafe, and unaccountable.

Roles of Medical Records in Healthcare:

#	Role	How it helps
1	**Clinical Care**	Doctors use past records to know the patient's history, allergies, and previous treatments — so they give the right treatment.
2	**Continuity of Care**	When a patient sees different doctors or hospitals, the record ensures consistent, safe treatment without repeating tests.
3	**Communication**	Doctors, nurses, lab staff, and pharmacists all refer to the same record — ensuring everyone is on the same page.
4	**Legal Protection**	Acts as evidence in court cases, insurance disputes, and medico-legal cases.
5	**Research & Education**	Medical students and researchers use anonymized records to study diseases, treatments, and outcomes.
6	**Billing & Insurance**	ICD codes from records are used to generate accurate bills and process insurance claims.
7	**Quality Assurance**	Medical audits use records to check if care meets standards and identify areas for improvement.
8	**Public Health & Statistics**	Disease outbreaks, death rates, birth rates — all tracked through medical records for government planning.

Flow Chart — Journey of a Medical Record:

🏥 Patient Arrives at Hospital — Patient comes to OPD, Emergency, or for Admission

↓

📝 Registration — New patient → OPD/IP record created with unique ID number

↓

🩺 Clinical Consultation — Doctor records: History, Examination findings, Diagnosis, Treatment plan

↓

🔬 Investigations — Lab tests, X-rays, ECG reports added to the record

↓

💊 Treatment / Surgery — Progress notes, drug charts, operation notes, nursing notes recorded daily

↓

🏠 Discharge — Discharge summary prepared — diagnosis, treatment given, follow-up advice

↓

📂 **MRD — Assembling**	All loose documents collected and arranged in standard order
↓	
🔍 **Deficiency Checking**	Missing entries identified and sent back to doctors for completion
↓	
🏷️ **Coding (ICD)**	Diagnoses and procedures coded for billing, statistics, and insurance
↓	
🗄️ **Indexing & Filing**	Record permanently stored in MRD using serial/terminal digit filing system
↓	
🔄 **Retrieval & Use**	Record retrieved when needed for: Follow-up care | Legal purpose | Research | Audit | Insurance

Conclusion: A medical record travels with the patient through every step of healthcare — from the moment they enter the hospital to long after they leave. It is the single most important document in healthcare delivery, connecting all departments and ensuring safe, accountable, and continuous care.